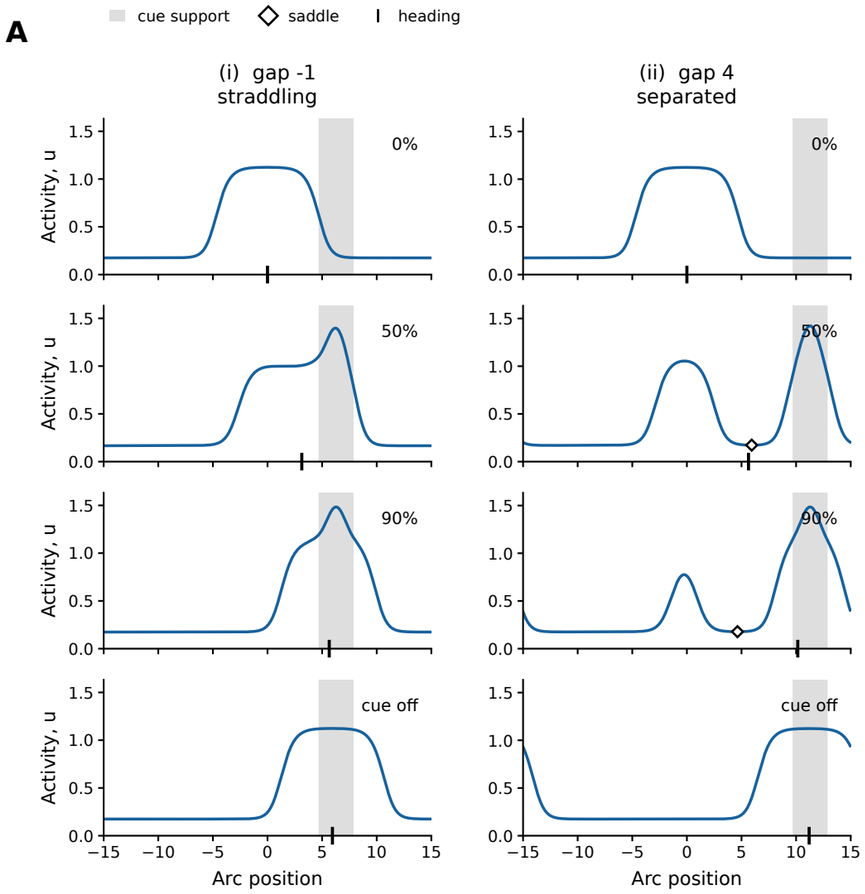
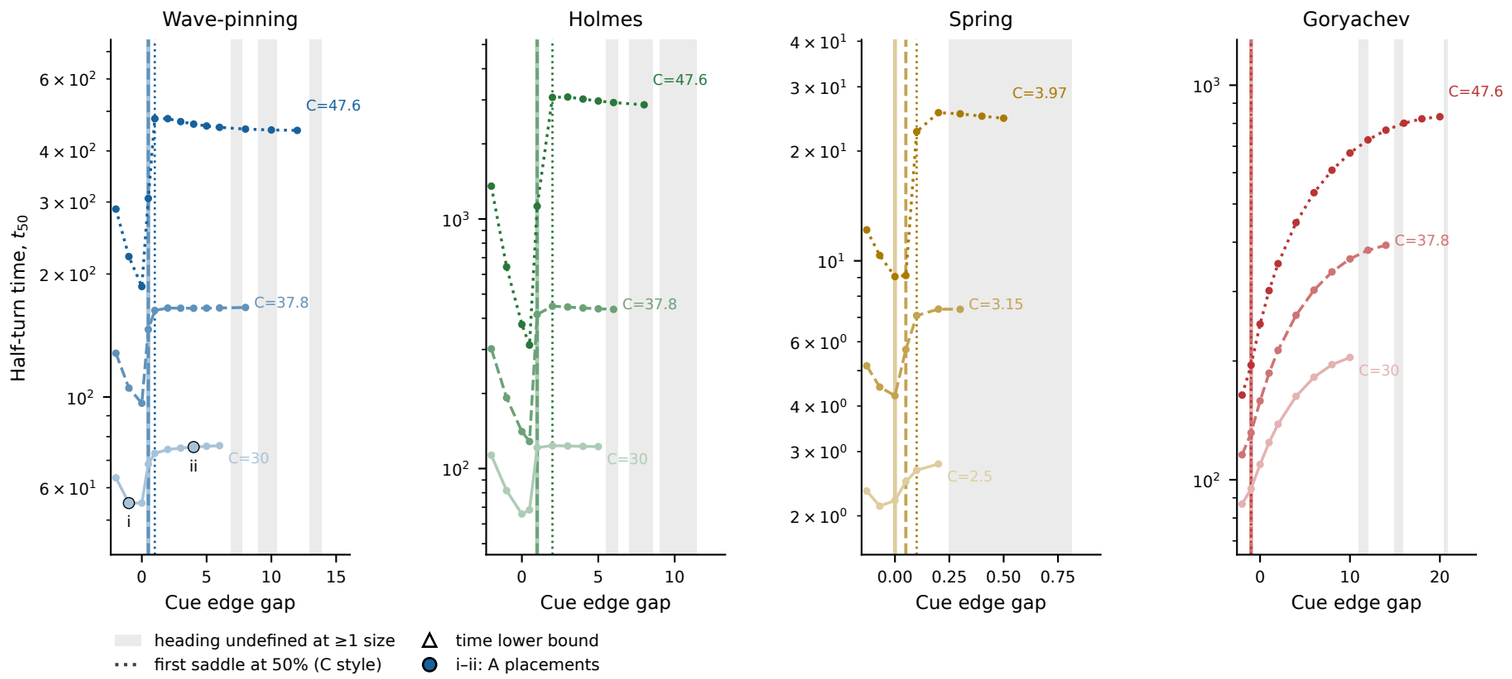
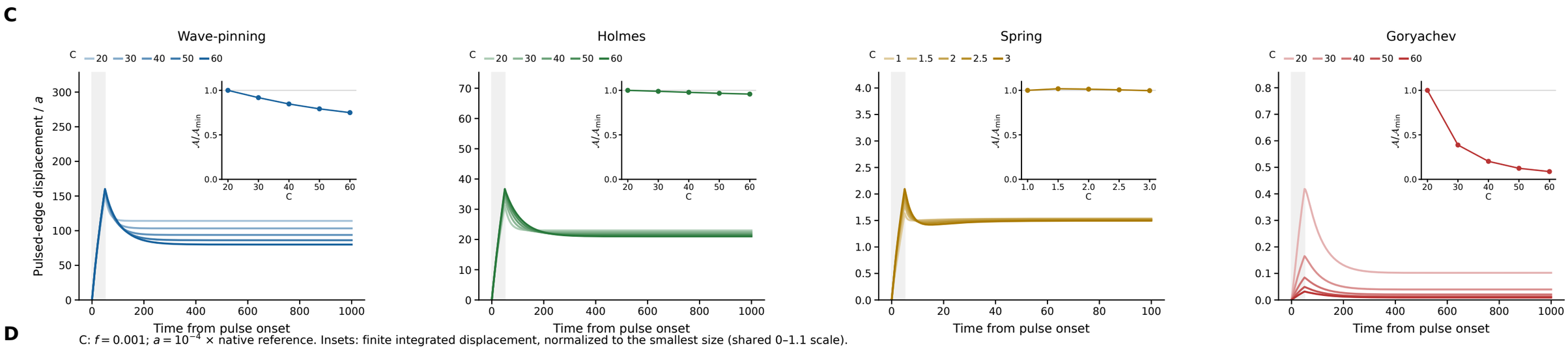


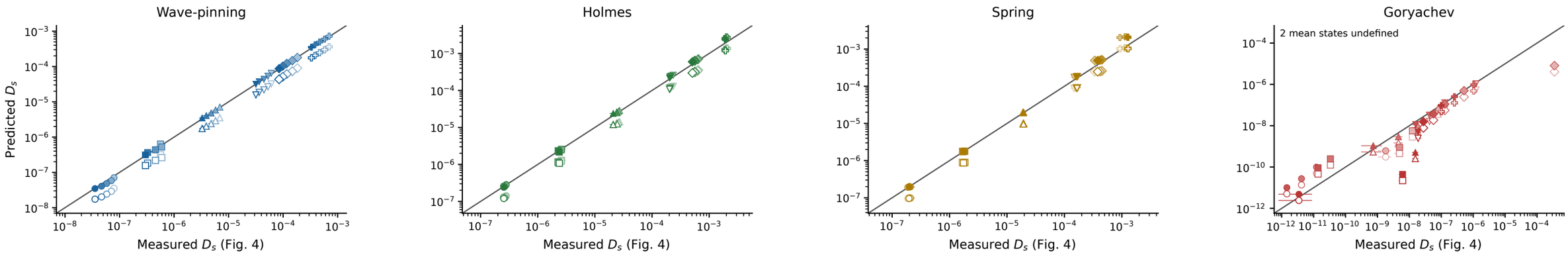


B: C = 30, 37.8, 47.6; Spring 2.5, 3.15, 3.97.
C–D: chemical C = 20–60; Spring 1–3. These are not matched size grids.



C: $f = 0.001$; $a = 10^{-4}$ × native reference. Insets: finite integrated displacement, normalized to the smallest size (shared 0–1.1 scale).

D



D: above, absolute diffusivity; below, size ratios at the same input. Lines: 1:1. Horizontal bars: whole-seed-bootstrap 95% CI.

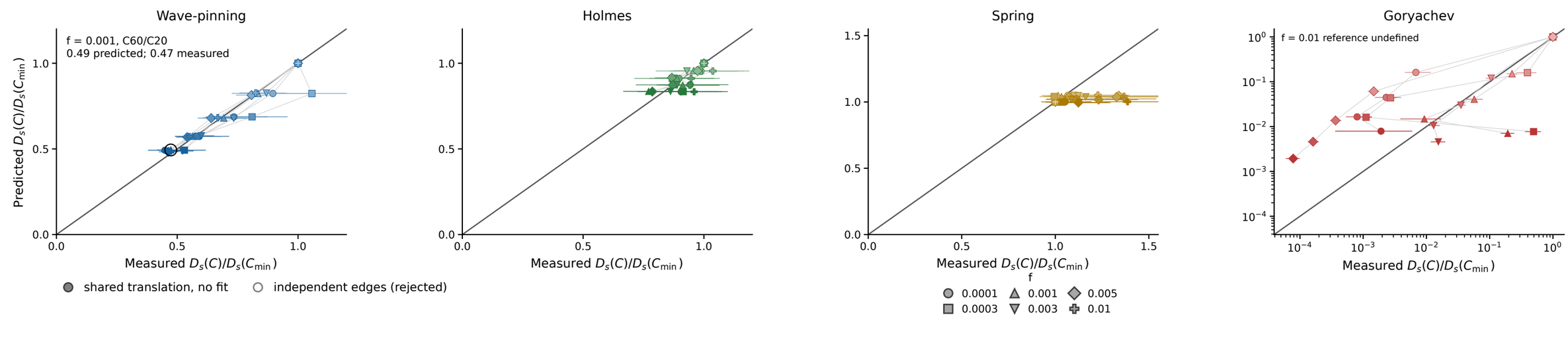


Figure 3. Enlargement slows both turning routes and selectively suppresses boundary susceptibility.

A. Wave-pinning activity at C = 30 for a cue straddling the incumbent boundary (i, gap −1) or separated from it (ii, gap 4). Rows show 0%, 50% and 90% of cue-direction heading, then the relaxed cue-free profile. Grey: cue support; diamonds: inter-region saddles; ticks: heading.

B. Half-turn time versus cue-edge gap from the incumbent's 10%-contrast boundary. Increasing sizes use lighter solid, intermediate dashed and darker dotted curves; chemical C = 30, 37.8, 47.6, Spring 2.5, 3.15, 3.97. Vertical lines mark the first resolved two-region saddle at half-turn; grey bands mark undefined heading at ≥1 size; triangles are time lower bounds. Enlargement slows both turning routes. C–D use different size grids, so these panels do not establish a matched-size relation between strong-cue turning and weak-pulse response.

C. Control-subtracted displacement of the pulsed half-contrast boundary divided by peak input a, at f = 0.001. A Gaussian pulse has SD ℓ = 1.5 and duration τ = 50 (Spring 0.15 and 5); peak a = 10^{-4} of the native input reference. Shades denote chemical C = 20–60 or Spring resting C = 1–3. Insets show displacement integrated to 20τ, divided by pulse dose aτ, then normalized to the smallest size on a common 0–1.1 axis. Largest/smallest terminal gains are 0.701 Wave-pinning, 0.914 Holmes, 0.989 Spring and 0.0878 Goryachev. Both edges and half/double amplitudes are checked in the supplement.

D. Above: predicted versus Figure 4 physical tracked drift for f ≤ 0.01. Filled markers retain shared translation; open markers retain the rejected independent-edge candidate. Measured pulse gain sets response amplitude; the unchanged native linearization supplies spatial shape, and the exact input covariance supplies forcing statistics. Nothing is fitted to drift. Below: predicted versus measured D_s(C)/D_s(Cmin) at the same f, removing the dominant input-strength scaling. Wave-pinning at f = 0.001 gives C60/C20 = 0.49 predicted and 0.47 measured. Lines: 1:1; bars: whole-seed-bootstrap 95% CI. Goryachev C20/C30 at f = 0.01 have no polarized mean-state boundary; the normalized prediction at that f also lacks its C20 reference.

Wave-pinning, Holmes and Spring have shared-translation prediction/observation medians 1.021, 1.080 and 1.069. Resolved quarter-window edge correlations are 0.980, 0.983 and 0.997 in the two exemplar seeds, refuting edge independence. Goryachev retains large quantitative disagreement. Tracked drift is conditional on trackable domains and is distinct from total heading decay.

Two cellular assays follow: vary photoactivation offset from straddling to separated placement and measure half-turn time; separately apply a weak brief boundary pulse and track both boundaries through withdrawal. The second assay measures susceptibility and common translation; it does not uniquely assign Figure 2 mechanism classes. Native units are not calibrated across models.

$$D_s^{\mathrm{ind}} = \frac{D_L + D_R}{4}, \qquad D_s = \frac{D_L + D_R + 2D_{LR}}{4} \tag{3.1}$$

Open markers test independent edges; filled markers retain the untuned shared-translation covariance.

Figure 3. Enlargement slows both turning routes and selectively suppresses boundary susceptibility. · continued

$$\widehat{g}_{e,j} = \frac{M_e}{Q_e} g_j, \qquad M_e = \frac{\Delta x_e(20\tau)}{a\tau}, \quad Q_e = \sum_j g_j h_{e,j} \tag{3.2}$$

Measured terminal displacement per pulse dose supplies amplitude; the unchanged native response supplies spatial shape. The covariance derivation is in the methods and supplement.